

To: Professor Benjamin Shaw

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Subject: EME 109 LAB 3 Report - Compressible Flow Nozzle Experiment

Equations

$$P_{\text{crit ratio}} = \frac{\text{Absolute chamber } P_c}{\text{Absolute inlet } P_I}$$

$$V_{\text{exit}} = \frac{F}{\dot{m}}$$

$$V_{\text{exit}} = c_p * T_0 \left[1 - \frac{P_e}{P_0}^{\frac{k-1}{k}} \right]$$

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$P_b \equiv$ chamber pressure

$P_0 \equiv$ inlet pressure

$$\dot{m} = \rho AV$$

speed of sound $C = \sqrt{kRT}$

$$M_a = \frac{V}{C}$$

$$\frac{T_0}{T} = 1 + \frac{k-1}{2} M_a^2$$

$$\frac{P_0}{P} = \frac{T_0}{T}^{\frac{k}{k-1}} = \left(1 + \frac{k-1}{2} M_a^2 \right)^{\frac{k}{k-1}}$$

$$\frac{\rho_0}{\rho} = \frac{T_0}{T}^{\frac{1}{k-1}} = \left(1 + \frac{k-1}{2} M_a^2 \right)^{\frac{1}{k-1}}$$

Can assume ideal gas:

$$P = \rho RT$$

Choked flow occurs when:

$$\frac{P}{P_0} = P_{\text{crit ratio}}$$

$$\frac{T^*}{T_0} = 0.833$$

$$\frac{P^*}{P_0} = 0.528$$

$$\frac{\rho^*}{\rho_0} = 0.634$$

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Opening

Findings/Summary

Discussion

0.1 Requirements

table of experimental and theoretical mass flow rate nozzle 1	y/n
table of experimental and theoretical mass flow rate nozzle 3	y/n
plot of experimental and theoretical mass flow rate nozzle 1	y/n
plot of experimental and theoretical mass flow rate nozzle 3	y/n
flow choked?	y/n
table of force calibration data	y/n
plot of force calibration data	y/n
Hysteresis present	y/n
polynomial fit if force cal. data using $\text{lm}()$	y/n
table of the thrust force vs abs chamber pressure nozzle 1	y/n
plot of the thrust force vs abs chamber pressure nozzle 1	y/n
table of the thrust force vs abs chamber pressure nozzle 3	y/n
plot of the thrust force vs abs chamber pressure nozzle 3	y/n
table of theoretical thrust force F	—
a. for the critical pressure ratio	y/n
b. for the condition where a normal shock is predicted at exit	y/n
c. design pressure ratio	y/n
calculate experimental isentropic efficiency of nozzle 1 for crit pressure ratio	y/n
calculate experimental isentropic efficiency of nozzle 1 for crit pressure ratio	y/n
calculate experimental isentropic efficiency of nozzle 3 for crit pressure ratio	y/n
calculate experimental isentropic efficiency of nozzle 3 for crit pressure ratio	y/n

0.2 Theoretical mass flow rate

Governing eq:

$$\dot{m} = \rho AV$$

The mass flow rate of a converging nozzle can be determined at exit state where:

$$\rho = \frac{P_{\text{exit}}}{RT}$$

$$A = A_{\text{exit}}$$

$$V =$$

Results

Closing

1 Photos

F791 NOZZLE PERFORMANCE TEST UNIT

